Nptel Online Certification Course: Computer Vision
Assignment 5: Stereo geometry
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type:MCQ**

Compute the essential matrix E, provided valid upper-triangular calibration matrices of two cameras in a stereo setup as:

$$K = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad K' = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

along with a valid rank-2 fundamental matrix F:

$$F = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$$

a) $\begin{bmatrix} 0 & 3 & 4 \\ 4 & 5 & 12 \\ -2 & -4 & 4 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 8 & 11 \\ 1 & 5 & 8 \\ 0 & 1 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 2 & 8 & 11 \\ 1 & 5 & 8 \\ -2 & -5 & -3 \end{bmatrix}$

d) None of the above

Correct Answer: a)
Detailed Solution:
Essential matrix $E = K'^T F K$.

QUESTION 2: **Type: MCQ**

Compute the homography induced by the plane at infinity, H_∞, for camera matrices

$$P = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \text{and} \quad P' = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

(a) $\begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Correct Answer: a)

Detailed Solution:

The homography induced by the plane at infinity is

$$H_\infty = M' M^{-1},$$

where M and M' are the left 3×3 sub-matrices of P and P' respectively:

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad M' = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since M is upper-triangular with ones on the diagonal,

$$M^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$H_\infty = M' M^{-1} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

QUESTION 3: **Type: MCQ**

Consider a stereo set-up with $P = \begin{bmatrix} 8 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and $P' = \begin{bmatrix} 8 & 0 & 0 & 16 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. If the image coordinates of a 3D point are $(2, 4)$ and $(6, 4)$ in left and right cameras respectively, compute its depth.

a) 2

b) 4

c) 6

d) 8

Correct Answer: b) **Detailed Solution:**
$P = K[I|0]$ and $P' = K[I|t]$,

$$K = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$t = \begin{bmatrix} 16/8 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

Using disparity relation, $x' = x + \frac{Kt}{Z}$

QUESTION 4: **Type: MCQ**

Suppose the motion of the cameras is pure translation with no rotation and no change in internal camera parameters. Let the epipole in the right image plane be $e' = (0, 0, 1)^T$. Compute the corresponding fundamental matrix.

(a) $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Correct Answer: (a)

Detailed Solution:

For pure translation without rotation,

$$F = [e']_\times$$

Given

$$e' = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

the skew-symmetric matrix is

$$[e']_\times = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

QUESTION 5: **Type: MCQ**

Given camera matrix $P = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$, compute the camera center C.

(a) $\begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} -2 \\ -3 \\ -4 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Correct Answer: (b)
Detailed Solution:
Camera center satisfies

$$PC = 0$$

For camera matrix $P = [M|p_4]$, camera center is

$$C = \begin{bmatrix} -M^{-1}p_4 \\ 1 \end{bmatrix}$$

Here, $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $p_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$.
Since $M^{-1} = I$,

$$-M^{-1}p_4 = \begin{bmatrix} -2 \\ -3 \\ -4 \end{bmatrix}$$

Thus,

$$C = \begin{bmatrix} -2 \\ -3 \\ -4 \\ 1 \end{bmatrix}$$

QUESTION 6: **Type: MCQ**

Given $P = \begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 4 & 2 & 2 \\ 3 & 1 & 1 & 1 \end{bmatrix}$, pseudo inverse $P^{+} = \begin{bmatrix} 0.2 & 0.1 & 0 \\ 0.1 & 0.2 & 0.1 \\ 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.1 \end{bmatrix}$, and camera center $C = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$,

Identify two points on the ray corresponding to image point $\mathbf{x} = (2,\ 1,\ 1)^T$.

(a) Only $P^{+}\mathbf{x}$

(b) C at $\lambda = 0$, $\quad P^{+}\mathbf{x}$ at $\lambda = \infty$

(c) $P^{+}\mathbf{x}$ at $\lambda = 0$, $\quad C$ at $\lambda = \infty$

(d) None of the above

Correct Answer: c)
Detailed Solution:
The back-projected ray is

$$X(\lambda) = P^{+}\mathbf{x} + \lambda C.$$

Thus the two points on the ray are $P^{+}\mathbf{x}$ at $\lambda = 0$ and camera center C at $\lambda \to \infty$. Note that $\mathbf{x}$ must be expressed in homogeneous coordinates, i.e. $\mathbf{x} = (2,\ 1,\ 1)^T$, so that $P^{+}\mathbf{x}$ is a valid 4×1 world point.

QUESTION 7: **Type:MCQ**

Suppose camera motion is pure translation parallel to y-axis with no rotation and no change in internal parameters. If epipole is $e' = (0, 1, 0)^T$, compute fundamental matrix.

(a) $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Correct Answer: b)
Detailed Solution:
Fundamental matrix is $F = [e']_{\times}$ and $e' = (0, 1, 0)^T$,

$$F = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

QUESTION 8: **Type:MCQ**

Consider stereo setup with $P = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and $P' = \begin{bmatrix} 6 & 0 & 0 & 12 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. If image coordinates are $(3, 5)$ and $(7, 5)$, compute depth.

(a) 2

(b) 3

(c) 4

(d) 5

Correct Answer: (b)
Detailed Solution:

$$P = K[I|0], \quad P' = K[I|t]$$

$K = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $t = \begin{bmatrix} 12/6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ Using disparity relation, $7 = 3 + \frac{6 \times 2}{Z}$

QUESTION 9: **Type:MCQ**

Check whether fundamental matrix $F = \begin{bmatrix} 2 & 1 & 0 \\ 3 & -1 & 2 \\ 1 & 2 & -3 \end{bmatrix}$ and camera matrices $P = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 1 & 1 & 2 \\ 3 & 2 & 1 & 1 \end{bmatrix}$, $P' = \begin{bmatrix} 2 & 1 & 3 & 2 \\ 1 & 2 & 2 & 1 \\ 3 & 1 & 1 & 2 \end{bmatrix}$ are compatible.

(a) Yes

(b) No

Correct Answer: (b)
Detailed Solution:
Compatibility condition requires

$$P'^T F P$$

to be skew symmetric.
Since computed matrix is not skew symmetric, matrices are not compatible.

QUESTION 10: **Type: MSQ**

Consider a stereo setup with camera matrices $P = K[I|0]$ and $P' = K[I|t]$, where translation is only along the x-axis and there is no rotation between the cameras. Which of the following statements are correct?

(a) The fundamental matrix must always be a full rank matrix.

(b) The epipolar lines in the two images are parallel.

(c) The essential matrix has rank 3.

(d) The disparity between corresponding image points can be used to estimate depth.

Correct Answer: (b), (d)

Detailed Solution:

For pure translation along x-axis with no rotation, the epipolar lines become parallel horizontal lines. Hence statement (b) is correct.

Fundamental matrix always has rank 2, not full rank. Hence statement (a) is incorrect.

Essential matrix also has rank 2 with one singular value equal to zero. Hence statement (c) is incorrect.

Depth can be estimated using disparity relation

$$Z = \frac{fB}{d}$$

where f is focal length, B is baseline, and d is disparity. Hence statement (d) is correct.
